

Wenbo Ji

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Homepage Google Scholar GitHub

3D/4D Scene Representations Controllable Video / World Models

Education

Technical University of Munich

M.Sc. Electrical Engineering and Information Technology

Double degree with Tongji University

Oct 2023 – Expected Nov 2026

Grade: 1.7 (1.0 = best)

Tongji University

M.Sc. Computer Science

Sept 2021 – Aug 2025

GPA: 4.5/5.0

Nanjing Tech University

B.Sc. Information and Computing Science

Sept 2017 – June 2021

GPA: 3.5/4.0

Selected Publications

* *Equal contribution* † *Corresponding author*

Video Diffusion Shader using 3D Face Tracking

2026

Wenbo Ji, Davide Davoli, Zhe Chen, Liam Schoneveld, Matthias Nießner, Jiapeng Tang†

In submission

Video diffusion shader conditioned on 3D face tracking for controllable one-shot head-avatar synthesis.

CSG-Fusion: Consistent Sparse-View Gaussian Splatting via Matching-based Fusion

2025

Yan Xia*†, **Wenbo Ji***, Weirong Chen, Daniel Cremers

ICCV 2025 Workshop on E2E3D – Best Paper Award

Matching-based fusion of sparse-view pointmaps into compact, cross-view-consistent 3D Gaussians.

LiteTracker: Leveraging Temporal Causality for Accurate Low-latency Tissue Tracking

2025

Mert Asim Karaoglu, **Wenbo Ji**, Ahmed Abbas, Nassir Navab, Benjamin Busam, Alexander Ladikos†

MICCAI 2025

Causal temporal feature reuse with prior-motion initialization for accurate, low-latency online tissue tracking.

RE0: Recognize Everything with 3D Zero-shot Instance Segmentation

2025

Xiaohan Yan*, Zijian Jiang*, Yinghao Shuai*, Nan Wang, Xiaowei Song,

Wenbo Ji, Ge Wu, Jinyu He, Gang Wei, Zhicheng Wang†

ICRA 2025

Training-free 3D zero-shot instance segmentation from multi-view masks and CLIP semantics.

Research Experience

Video-Action World Models for Robot Dexterous Manipulation

Apr 2026 – Present

Agile Robots SE, World Models & Dexterous Manipulation

Research Intern

Mentors: Mahdi Mustapha Hamad

- Proposing a video-generation method that translates human egocentric manipulation videos into robot-domain videos to improve robot policy learning.

Human Motion Video Diffusion

Mar 2026 – Present

TUM Visual Computing Group

Master's Thesis

Advisor: Prof. Matthias Nießner; Co-advisors: Yu Chi, Jiapeng Tang

- Developing camera-controlled human motion video diffusion for controllable synthesis of human-scene interactions.

Video Diffusion Shader (Head Avatar)

Apr 2025 – Mar 2026

TUM Visual Computing Group

Research Intern

Mentors: Jiapeng Tang, Prof. Matthias Nießner

- Developed Video Diffusion Shader using 3D Face Tracking, a one-shot head-avatar video pipeline conditioned on 3D face tracking.

LiteTracker: Online Tissue Tracking

Aug 2024 – Apr 2025

ImFusion | TUM CAMP

Research Intern

Mentors: Mert Asim Karaoglu, Prof. Benjamin Busam, Prof. Nassir Navab, Alexander Ladikos

- Implemented online inference and EMA-flow initialization for LiteTracker (MICCAI 2025), and ran the training experiments.
- Achieved competitive tracking on STIR and SuPer with 29.67 ms P95 latency for 1,024 points on an NVIDIA RTX 3090.

CSG-Fusion: Sparse-View Gaussian Splatting

Feb 2024 – Aug 2025

TUM CVG | Oxford VGG

Guided Research

Mentors: Prof. Daniel Cremers, Prof. Yan Xia, Weirong Chen, Prof. Chuanxia Zheng

- Proposed and developed CSG-Fusion end-to-end (method idea and all experiments): matching-based pruning, mask-based feature aggregation, and cross-view supervision.
- On ScanNet++, +2.8 dB PSNR over Splatt3R at 90% overlap with $\sim 124K$ fewer Gaussians; also evaluated zero-shot on DTU.

Endoscopic Scene Reconstruction with 4D Half Gaussian Splatting

2025

Tongji University

Master's Thesis

- Developed a 4D Half-Gaussian splatting pipeline for deformable stereo endoscopic reconstruction with depth-prior initialization, HexPlane spatio-temporal deformation, and edge-aware depth regularization.
- Achieved 38.1 PSNR on EndoNeRF versus prior endoscopic GS/NeRF baselines (e.g., EndoGaussian, Deform3D-GS); also evaluated on SCARED.

Honors

DAAD Scholarship

2023 – 2024

National Encouragement Scholarship (China)

2019